Where Workers Come From: An Occupational Flow Account for the United States, and the Instrument That Makes It Measurable

Alessandro Conway

2nd September 2026

Extended summary

Across the occupations the Current Population Survey can support, arrivals from another occupation run between four and twelve times arrivals from education. The ratio holds for 98 per cent of occupations and survives aggregation to the twenty three major groups, where the classification cannot confuse a nurse with a physical therapist, and the arrivals are spread across the age distribution: for registered nurses the rate runs 3.1, 2.2, 3.4, 1.3 and 2.8 per hundred from age twenty five upwards, so they are not graduates recorded under the wrong heading.

Getting to that point required settling a measurement question first. Occupational mobility is usually measured by following a person across two interviews. The survey assigns occupation codes from written descriptions of what a person does, and two independent codings of the same unchanged job disagree in 48.1 per cent of cases at the detailed level and 31.8 per cent across the twenty three major groups. The figure comes from 114,491 people whose occupation the survey records as unchanged and who worked all fifty two weeks of the year. State of residence and sex disagree in 0.00 per cent of those cases; education disagrees in 12.6. A linked measure of annual mobility returns 50.0 per cent, below the disagreement between two codings of a period during which nothing could have changed.

The March supplement asks in one interview what a person does now and what they did last year. Both codes are assigned at one sitting from one respondent's description, their errors are correlated, and the correlation cancels in the difference. That is why an instrument relying on recall outperforms a panel.

The account itself covers 230 occupations, annually from 2009 to 2025, by age band, with a standard error from the 160 replicate weights the Census Bureau publishes. Registered nurses have the lowest total departure rate among the large health and education occupations, at 9.4 per hundred a year, while nursing and home health aides lose 22.7, most of it to other occupations. Most movement between occupations also cancels: only 36.4 per cent of gross lateral flow carries direction at the detailed level, 22.1 at minor group and 14.5 at major group, so an occupation is churned by flows several times larger than the change in its size.

1 Introduction

An occupation that cannot find enough people is usually treated as an occupation that has not trained enough of them, and states fund nursing school places, teacher preparation, apprenticeships and community college capacity on that understanding. The diagnosis has rarely been checked against a full account of where an occupation's people come from, because no such account exists at the level of detail policy operates on: employment statistics report stocks with precision and at high frequency, they say nothing about the flows that produced them, and an occupation stable in headcount is therefore recorded identically to one replacing a fifth of its people every year.

Flows themselves have been studied for decades. Work on gross worker flows established that net employment change is the small residual of much larger movements between employment, unemployment and inactivity (Davis and Haltiwanger, 2014). Work on occupational mobility established that people change occupation often enough for occupation specific human capital to matter for wages (Kambourov and Manovskii, 2008). Recent work treats transitions between occupations as a network with its own structure (Cheng and Park, 2020). None of it produces a table a workforce planner could open and read off, for a named occupation, how many people arrived from training, how many arrived from another occupation, how many left for another occupation and how many left work altogether.

Assembling one requires settling a measurement question the mobility literature has circled for two decades. Kambourov and Manovskii (2008) devoted substantial effort to spurious transitions arising from independent coding, and Mellow and Sider (1983) showed much earlier that worker and employer reports of the same job frequently disagree. What has not been available is a direct measurement of the problem, on people whose occupation is known not to have changed. Section 2 provides one, Section 3 builds the account, Section 4 reports it, and Section 5 sets out the scope of the measures. A companion paper takes the same account to the geography and to the interstate movement of workers by occupation, and nothing here depends on it.

2 The instrument

2.1 Three measures of one transition

The Current Population Survey records occupation by asking what kind of work a person does, writing down the answer, and coding the description. Its March supplement asks that question twice, once about the present and once about the longest job held during the previous calendar year. The difference between the two codes is the annual transition used here.

The alternative is to follow a person across interviews, and the rotation design returns a household to the sample twelve months after its first interview, so the same person appears in March of one year and March of the next, with occupation collected at the time on both occasions and an eight month gap during which no answer is carried forward. Linking the supplement to the March monthly file for the same month gives agreement on current occupation in 99.99 per cent of 504,531 people, so the two files share one coding of the present job. Table 1 sets those two measures against a third: take last year's occupation as recorded contemporaneously a year earlier, compare it with last year's occupation as recalled in the supplement, and although both describe the same period they disagree 52.2 per cent of the time.

Table 1: Three measures of the same annual transition, 2017 to 2025

	Detailed (474 codes)	Major group (23)
Same period, two independent codings	52.2%	33.5%
Twelve months apart, two independent codings	50.0%	
Twelve months apart, coded in one interview	10.3%	3.3%

The second row follows from the first. A linked measure of annual mobility returns 50.0 per cent, below the disagreement over a period in which nothing could have changed, so a linked panel applied to this survey recovers coding discordance with real movement somewhere underneath and no way to separate the two.

2.2 Which instrument is wrong

A disagreement establishes that one of two measurements is wrong without saying which, and two codes assigned at one sitting from one description could take the same value out of consistency alone.

Restrict attention to people whom the supplement records as having held the same occupation in both years and who worked all fifty two weeks. For these 114,491 people the independently coded occupation from a year earlier disagrees in 48.1 per cent of cases at the detailed level, 41.1 across the ninety seven minor groups and 31.8 across the twenty three major groups. Figure 1 sets that against the movement the supplement measures on the same people.

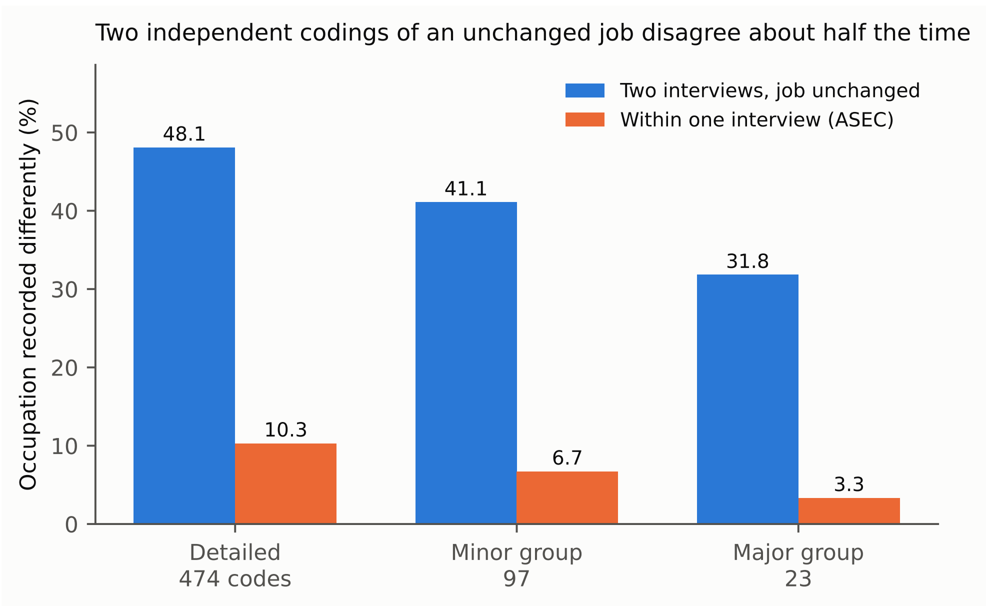

Figure 1: Two independent codings of an unchanged job disagree about half the time. The within interview measure records annual movement an order of magnitude smaller.

Three things point at the coding and not at the supplement. The person link holds: state of residence differs in 0.00 per cent of cases and sex in 0.00, education in 12.6 and occupation in 48.1, making occupation the least reliable variable in the survey by some distance. The disagreement sits near fifty per cent in all nine year pairs from 2016, including those either side of the 2020 change of occupation classification. The third concerns the shape of the disagreements, which would look like career moves if the supplement were suppressing real ones, and do not: the largest single confusion is elementary and middle school teachers against secondary school teachers, at 0.96 per cent of all discordance, followed by first line supervisors against the staff they supervise in retail and in offices, by miscellaneous managers against specific managerial categories, by nursing aides against personal care aides, and by accountants against bookkeepers. Each is a boundary in the classification where a description could be coded either way.

The test bounds the disagreement between two independent codings of one job. It does not measure how far a single interview suppresses real change through the consistency of two codes assigned at one sitting.

Discordance concentrates within occupational families and along the supervisory boundary. Movement between families survives it; movement within a family does not. Registered nurses against physical therapists, a pair whose flow in the account looked implausible to me before this test, accounts for 0.002 per cent of discordance. Elementary against secondary teachers is the worst pair in the data and should be reported only at a level of aggregation containing both.

3 The flow account

3.1 Construction

The account uses the seventeen annual supplements from 2009 to 2025, comprising 2,924,246 person records. For each occupation and year it records the number of people who held the occupation during the previous year, the number arriving from education, from other non-employment and from another occupation, and the number departing to another occupation or out of employment. Arrivals from outside employment carry the previous year occupation code reserved for people who did not work, and the stated reason for not working separates education from caring, illness, retirement and inability to find work.

The identity closes to zero people, since every person contributes once to each side from one source and no term is calibrated to an external total.

Two construction details matter for reading the figures. The 2014 supplement was fielded as a split sample, three eighths of respondents receiving redesigned income questions and five eighths the existing ones, so analysing both halves together returns weighted totals twice the size of the population; filtering on the flag that marks the halves gives 144.8 and 145.5 million against 290.3 million unfiltered, and both halves sit on the trend between the adjacent years. Standard errors come from the 160 person replicate weights the Census Bureau constructs by successive difference replication, with the multiplier of 4/160 the documentation prescribes.

Rates are recomputed inside each replicate, since numerator and denominator move together.

3.2 Coverage and precision

Precision degrades quickly as occupations get smaller. The median standard error on a rate is 1.3 percentage points where at least three hundred prior year holders are sampled, 2.8 points between one hundred and three hundred, 4.8 points between fifty and one hundred, and 10.6 points below fifty. A cell is reported only where at least one hundred prior year holders appear, a threshold set from that relationship before the estimates were inspected. That leaves 230 occupations across the full period and 172 for 2022 to 2025, out of the 474 the survey distinguishes.

4 What the account shows

4.1 Arrivals

Table 2 gives the account for eight occupations, averaged over 2022 to 2025 and expressed per hundred people who held the occupation the previous year. Figure 2 shows arrivals and departures on a common scale.

Table 2: Arrivals and departures per 100 prior-year holders, average of 2022 to 2025

	Arrivals from			Departures to	
Occupation	Education	Other non-work	Another occupation	Another occupation	Out of work
Registered nurses	0.7	0.7	3.4	4.4	4.9
Elementary and middle teachers	0.5	1.1	3.6	4.9	5.4
Secondary teachers	1.0	0.6	7.6	6.7	4.7
Software developers	0.7	0.5	4.1	6.7	5.5
Truck drivers	0.4	2.1	7.7	9.3	9.6
Nursing and home health aides	0.9	1.8	13.1	13.2	9.5
Cashiers	4.8	3.0	8.7	12.3	20.3

Two features of the table cut against the way workforce shortage is discussed. Registered nurses have the lowest total departure rate among the large health and education occupations, 9.4 per hundred a year; nursing and home health aides lose 22.7, and the gap between them sits in movement to other occupations, so what is discussed as a nursing shortage appears in the flows as a retention problem in the occupations next to nursing. The pattern extends well beyond these eight occupations. Figure 3 plots every occupation in the account, and 98 per cent sit above the line where arrivals from other occupations equal arrivals from outside employment. Across the whole matrix, arrivals from another occupation run 4.29 times arrivals from outside employment at the detailed level, 2.89 at minor group and 1.55 across the twenty three major groups, where the classification cannot confuse a nurse with a physical therapist and the surviving flow is movement between broad occupational families. Against arrivals from education alone, the ratio is 4.3 even at that coarsest level.

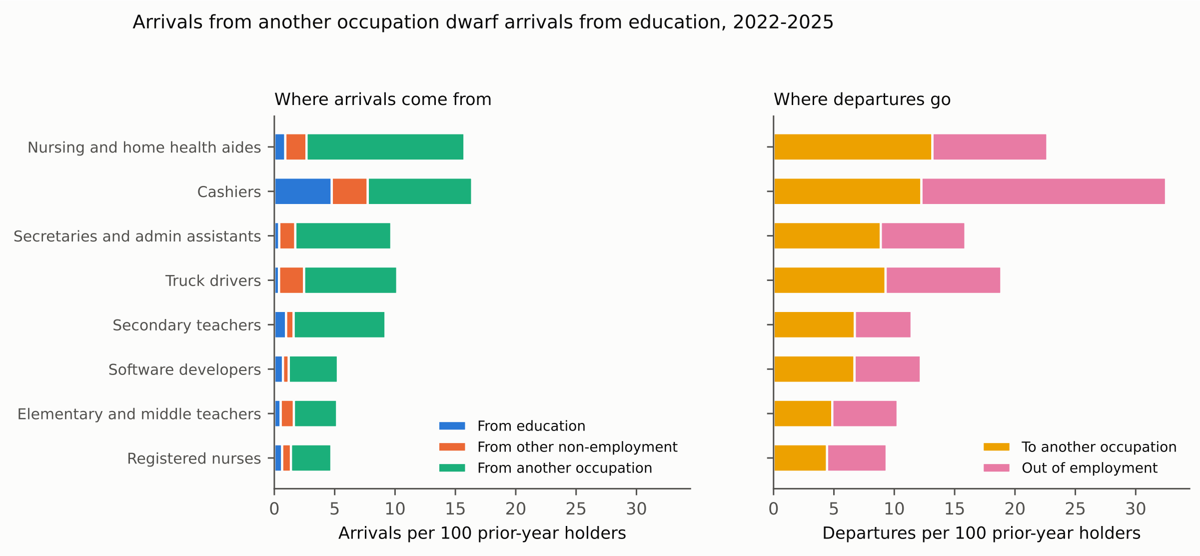

Figure 2: Arrivals from another occupation dominate arrivals from education, and the ranking by turnover runs against the way shortage is usually described.

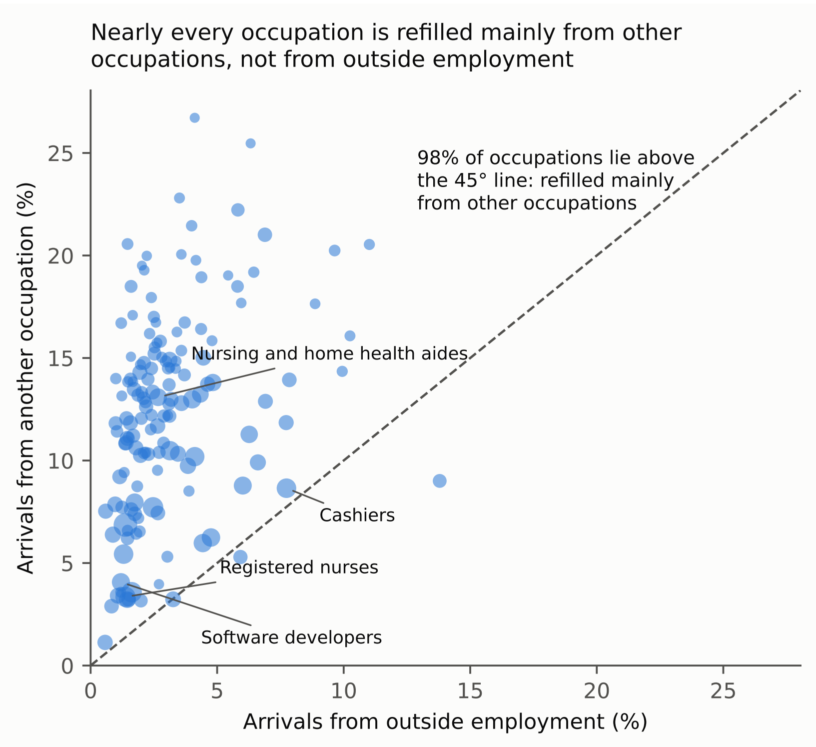

Figure 3: Arrivals from another occupation against arrivals from outside employment, one point per occupation, 2022 to 2025.

4.2 The age profile

Arrivals recorded as coming from another occupation might be graduates who held a job while studying, which would put the training pipeline back in the account under a different heading and would concentrate such arrivals in the youngest bands. Figure 4 shows arrivals from another occupation by age for three occupations, and for registered nurses the rate runs 3.1, 2.2, 3.4, 1.3 and 2.8 per hundred across the bands from twenty five upwards. Between 42 and 55 per cent of lateral arrivals land in the two youngest bands, against roughly a third of the workforce being that age.

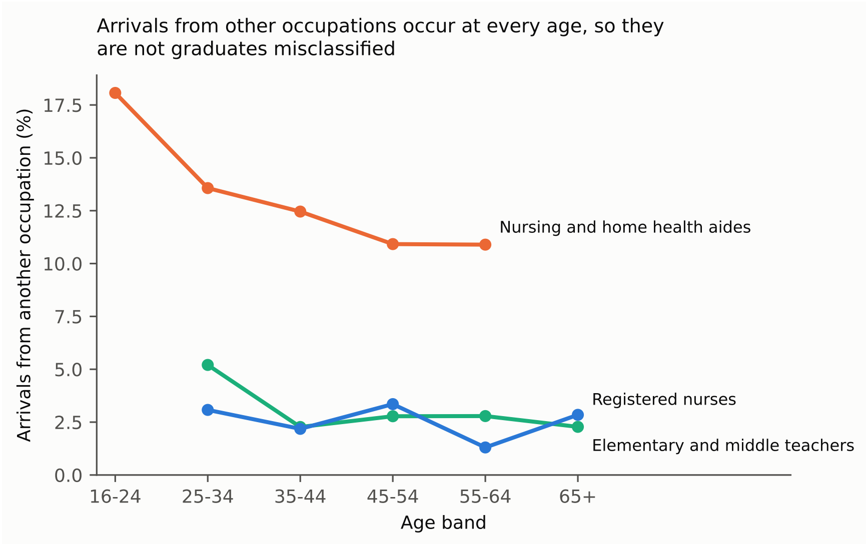

Figure 4: Arrivals from another occupation occur at every age.

4.3 Direction and churn

Gross movement between occupations is large and most of it cancels. Splitting the transition matrix into a symmetric part and an antisymmetric one separates two way traffic between a pair of occupations from net reallocation between them, and symmetric misclassification cannot produce the second. At the detailed level, gross lateral movement of 18.81 million a year contains 3.43 million of net directional reallocation, a directional share of 36.4 per cent. At minor group the figures are 12.67 and 1.40 million, a share of 22.1 per cent; at major group 6.80 and 0.49 million, a share of 14.5 per cent.

The symmetric remainder is bounded and not identified, since two way movement between adjacent occupations and coding discordance both live there. What the decomposition establishes is a floor on real reallocation, and the size of that floor against the gross flow is the point: an occupation's stock is churned by movements several times larger than the change in its size, and a policy raising arrivals works against departures of comparable magnitude.

5 Scope of the measures

Three features of the design bear on how the figures should be read. The count of people who held an occupation last year covers a twelve month window while the count of current workers is taken at a single moment, so the ratio between the two is not a growth rate, though the flows and comparisons between occupations are unaffected. The survey looks back one year, so arrivals from outside employment include people returning to an occupation they had held before alongside those entering it for the first time. And a departure is the statement that a person was not in the occupation twelve months later, which a one year transition cannot separate from a temporary absence.

6 Conclusion

Occupations in the United States are refilled predominantly by people arriving from other occupations, at four to twelve times the rate at which they are refilled from education. The ratio holds for 98 per cent of occupations and survives aggregation to a level at which the classification cannot confuse adjacent professions, the arrivals occur at every age, and most of the movement cancels, leaving gross flows several times the net change in an occupation's size.

Two independent codings of an unchanged job disagree in half of cases, so occupational mobility cannot be measured in this survey by following people between interviews, and the retrospective design, whose reliance on recall makes it look the weaker instrument, is the one that survives the comparison.

Training capacity acts on the smallest of an occupation's inflows. The largest is governed by what determines who can move into an occupation from another one: credential requirements, licensure, and the transferability of skills between adjacent occupations. Which of those binds, and for which occupations, is the question the account was built to make askable.

References

Cheng, S. and Park, B. (2020) 'Flows and boundaries: a network approach to studying occupational mobility in the labor market', *American Journal of Sociology*, 126(3), pp. 577–631.

Davis, S. J. and Haltiwanger, J. (2014) 'Labor market fluidity and economic performance', *NBER Working Paper* 20479.

Flood, S., King, M., Rodgers, R., Ruggles, S., Warren, J. R. and Westberry, M. (2024) *Integrated Public Use Microdata Series, Current Population Survey*. Minneapolis: IPUMS.

Kambourov, G. and Manovskii, I. (2008) 'Rising occupational and industry mobility in the United States, 1968–97', *International Economic Review*, 49(1), pp. 41–79.

Mellow, W. and Sider, H. (1983) 'Accuracy of response in labor market surveys: evidence and implications', *Journal of Labor Economics*, 1(4), pp. 331–344.

Moscarini, G. and Thomsson, K. (2007) ‘Occupational and job mobility in the US’, *Scandinavian Journal of Economics*, 109(4), pp. 807–836.